



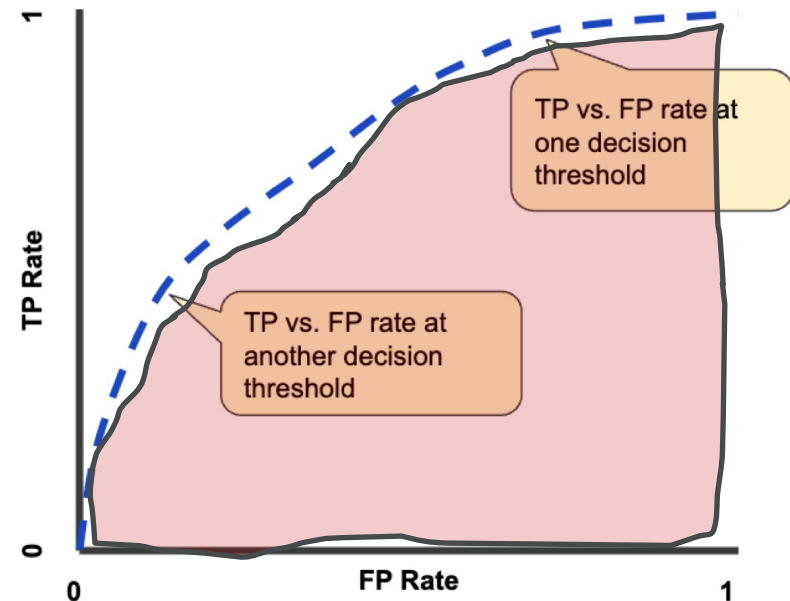
Project Introduction: Privacy Preserving Computation of AUC

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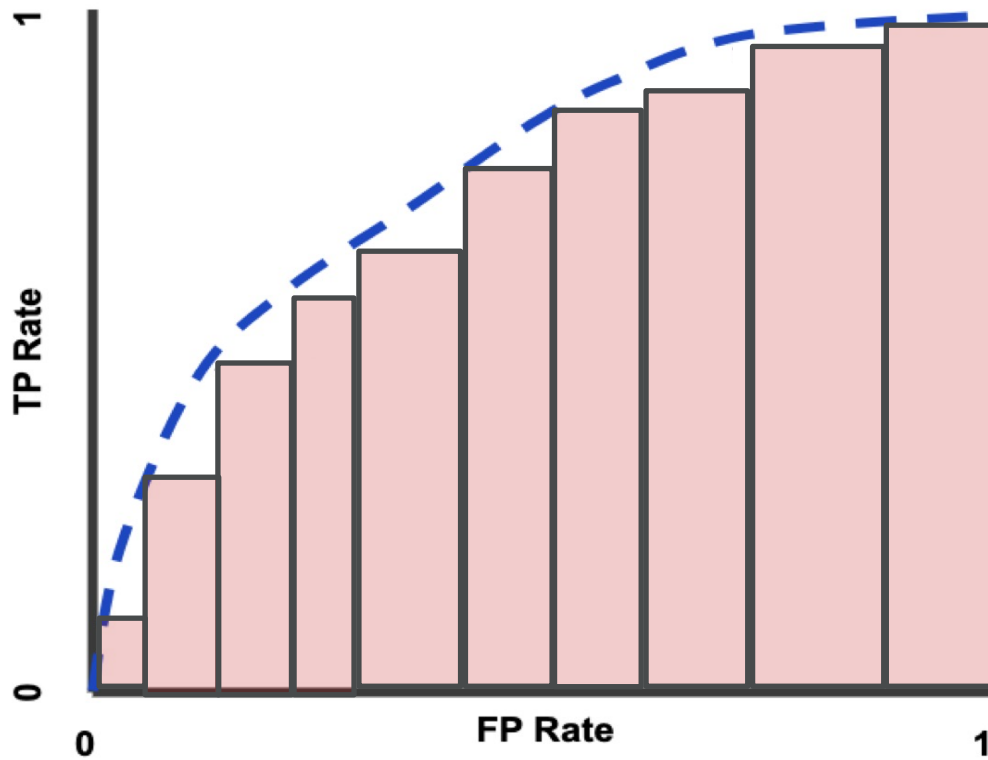
ROC and AUC

- Performance metric for binary classifier based on confidence scores
- **ROC** = Receiver operating characteristic curve: illustrates performance of binary classifier model at varying threshold values (classification thresholds)
- **AUC** = Area under the ROC curve



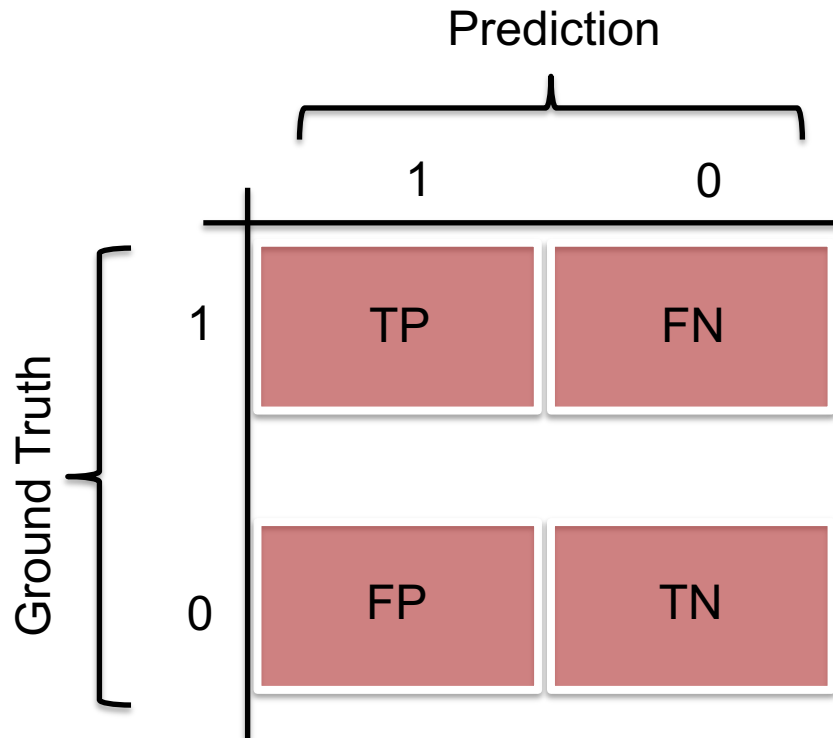


ROC and AUC





TPR and FPR



True positive rate:

$$TPR_{\theta} = \frac{TP_{\theta}}{TP_{\theta} + FN_{\theta}} = \frac{TP_{\theta}}{P}$$

False positive rate:

$$FPR_{\theta} = \frac{FP_{\theta}}{FP_{\theta} + TN_{\theta}} = \frac{FP_{\theta}}{N}$$

θ : classification threshold



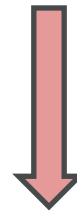
AUC

$$AUC = \sum_{i=1}^{|\tau|} \frac{(TPR_i + TPR_{i-1}) \cdot (FPR_i - FPR_{i-1})}{2}$$



AUC Computation

Prediction Confidence Scores:	0.5	0.9	0.1	0.3	0.7
Ground Truth Labels:	1	1	0	0	0



Sorting

Prediction Confidence Scores:	0.9	0.7	0.5	0.3	0.1
Ground Truth Labels:	1	0	1	0	0



AUC Computation

$$AUC = \sum_{i=1}^{|\tau|} \frac{(TPR_i + TPR_{i-1}) \cdot (FPR_i - FPR_{i-1})}{2}$$

Ground Truth	Score	Threshold						
		0	0.1	0.3	0.5	0.7	0.9	1.0
1	0.9							
0	0.7							
1	0.5							
0	0.3							
0	0.1							

TPR							
FPR							



AUC Computation

$$AUC = \sum_{i=1}^{|\tau|} \frac{(TPR_i + TPR_{i-1}) \cdot (FPR_i - FPR_{i-1})}{2}$$

Ground Truth	Score	Threshold						
		0	0.1	0.3	0.5	0.7	0.9	1.0
1	0.9	1	1	1	1	1	1	0
0	0.7	1	1	1	1	1	0	0
1	0.5	1	1	1	1	0	0	0
0	0.3	1	1	1	0	0	0	0
0	0.1	1	1	0	0	0	0	0

TPR	2/2	1	1	1	1/2	0.5	0
FPR	3/3	1	2/3	1/3	1/3	0	0